



UCF-439
Capstone Project

**AUTOSHIELD: A REAL-TIME MACHINE LEARNING-DRIVEN DDOS
DETECTION AND MITIGATION SYSTEM**

Group Details:

Group No. 71

Angela Paul SAP: 1000018001

Krishna Sharma SAP: 1000018720

Kushagra Upadhyaya SAP: 1000018770

Supervisor Details

Supervisor Name : **Prof. Mohd. Shariq**



Introduction

AutoShield is a comprehensive, real-time autonomous security platform designed to detect and mitigate Distributed Denial of Service (DDoS) attacks without human intervention. The project specifically addresses the fundamental inadequacy of traditional signature-based Intrusion Detection Systems (IDS), which cannot adapt to novel attack patterns and depend on slow, manual response cycles. By shifting from reactive manual processes to an automated, millisecond-latency defense model, the system bridges the critical gap between intelligent threat detection and active network defense.

The system's core architecture integrates high-level network engineering with sophisticated machine learning to convert raw packets into high-fidelity intelligence. Built on a three-layer pipeline, it leverages Apache Kafka for robust data ingestion and PySpark Structured Streaming for immediate telemetry analysis. When the analytical "brain"—powered by models like Random Forest—identifies a threat, it triggers an immediate "muscle" response via an automated firewall controller. This triggers Linux **iptables** rules within milliseconds to neutralize attacks at the source, achieving a closed-loop defense cycle that requires no operator involvement.



Background/Related Work

Background

- Modern cyberattacks such as DDoS occur at multiple layers (network, protocol, and application), making them difficult to detect and mitigate.
- Volumetric attacks (UDP/ICMP floods) overwhelm bandwidth, while protocol attacks (SYN floods) exhaust system resources.
- Application-layer attacks (HTTP floods, Slowloris) closely mimic real user traffic, making detection more complex.
- Threat intelligence data is scattered across multiple sources like IP reputation systems, CVE databases, and malware feeds.
- Combining and analyzing this data manually is time-consuming and inefficient for real-time protection.



Related Work

- Traditional tools like Snort and Suricata rely on signature-based detection and fail to detect zero-day or unknown attacks.
- Machine learning approaches (e.g., Random Forest on CIC-DDoS datasets) have shown high accuracy but are mostly limited to offline analysis.
- Deep learning models such as LSTM and CNN improve detection accuracy but suffer from high latency, making them less suitable for real-time systems.
- Existing automated mitigation systems use SDN or firewall APIs but lack integration with real-time threat intelligence and ML models
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- There is no unified open-source system that combines threat data aggregation, real-time ML classification, and automated firewall response.



Feasibility Study

Technical Feasibility

- The system is built using **Python 3.11**, which provides a strong ecosystem for data processing, machine learning, and automation.
- Real-time data ingestion is handled using **Kafka streaming**, enabling high-throughput processing of threat intelligence data.
- Machine learning is implemented using **Random Forest (scikit-learn)**, providing high accuracy (~99.85%) with low inference time (~1 ms per record).
- Data processing and storage are managed using **pandas, pyarrow (Parquet), and SQLite**, ensuring fast and efficient handling of large datasets.

Operational Feasibility

- The system runs on **standard Windows/Linux machines**, requiring no specialized hardware.
- Fully automated pipeline with a **scheduler (daily execution at 02:00 UTC)** reduces the need for manual monitoring.
- Demonstrated ability to process real-time data and generate firewall rules in **under 3 minutes**.
- Easy to maintain and extend due to modular architecture (ETL, ML, Firewall separated).
- Suitable for small organizations as well as scalable for larger systems.



Economic Feasibility

- The entire project is built using **open-source technologies**, resulting in zero licensing cost.
- No need for expensive commercial security solutions or hardware appliances.
- Can be deployed on **low-cost or existing infrastructure**, reducing overall investment.
- Automated system reduces human effort and operational cost in long-term maintenance.

Risk Assessment

- **False positives (Medium)**: Controlled using class balancing and threshold tuning in ML model.
- **API rate limits (Medium)**: Managed using delay and retry mechanisms.
- **Kafka load issues (Low)**: Handled through scalable partitioning.
- **New attack evasion (Low)**: Reduced by periodic retraining of the ML model.



Project Description

- **Autonomous Defense:** A self-contained, machine learning-driven Intrusion Prevention System (IPS) that monitors and mitigates malicious network flows without human oversight.
- **Modular Pipeline:** Built as a three-tier, closed-loop system consisting of an Ingestion Layer (packet capture), an Analysis Layer (intelligence), and a Reaction Layer (firewall control).
- **High-Speed Streaming:** Utilizes Apache Kafka and PySpark Structured Streaming to convert raw network packets into high-fidelity intelligence in real-time.
- **Intelligent Classification:** Employs optimized Random Forest and XGBoost models to distinguish malicious actors with approximately 99.85% accuracy.
- **Instant Mitigation:** Automatically updates the Linux kernel-level firewall (**iptables**) within milliseconds to block offending IP addresses at the source.
- **Real-Time Visibility:** Features a Streamlit dashboard providing live throughput analytics, threat geolocation, and "Explainable AI" heatmaps for security analysts.



Problem Statement

- Reliance on Reactive Defenses:** Traditional security tools like Snort and Suricata rely on static signatures and manual intervention, which results in significant damage before a human administrator can respond.
- Zero-Day Vulnerability:** Signature-based systems are unable to detect previously unseen attack variants or complex, multi-vector attacks that combine volumetric and protocol-layer tactics.
- Throughput and Scalability Bottlenecks:** Modern volumetric DDoS attacks generate millions of packets per second, which often exceed the processing capacity of single-threaded analyzers, leading to missed detections.
- Lack of Closed-Loop Response:** Existing open-source solutions typically provide discovery but not mitigation creating a dangerous time gap between identifying an attack and blocking it.
- High-Dimensional Complexity:** The sheer volume and dimensionality of network telemetry make it nearly impossible for human analysts to identify subtle anomalies in real-time without intelligent automation.



Proposed Solution – “Sandwich Architecture”

- **Six-Stage Pipeline:** The system operates as a six-stage closed-loop pipeline:
Ingestion → Buffering → Processing → Intelligence → Mitigation → Visualization.
- **Layered Separation:**

Ingestion Layer (The Senses)

- **Stage 1: Packet Ingestion:** Utilizing the **Scapy** library in promiscuous mode, the system acts as a sensory organ, sniffing every incoming packet at the Network Interface Card (NIC).
- **Stage 2: Buffering:** Raw packet metadata (source/destination IPs, protocols, and payload lengths) is serialized into JSON and published to **Apache Kafka**. This acts as a critical "shock absorber," allowing the system to ingest millions of events per second without data loss during massive volumetric spikes.



Analysis Layer (The Brain)

- **Stage 3: Processing & Feature Extraction:** **PySpark Structured Streaming** consumes the Kafka feed in 5-second micro-batches. It aggregates individual packets into "network flows" based on a **5-tuple key** (Source/Destination IP, Ports, and Protocol) while calculating over 20 behavioral metrics like packet rate and protocol entropy.
- **Stage 4: Intelligence & Classification:** The system's "analytical brain" applies an optimized **Random Forest model** to the extracted flow features. To maintain ultra-low latency, the model is distributed across Spark worker nodes using broadcast variables, achieving **99.85% accuracy** with near-zero inference time



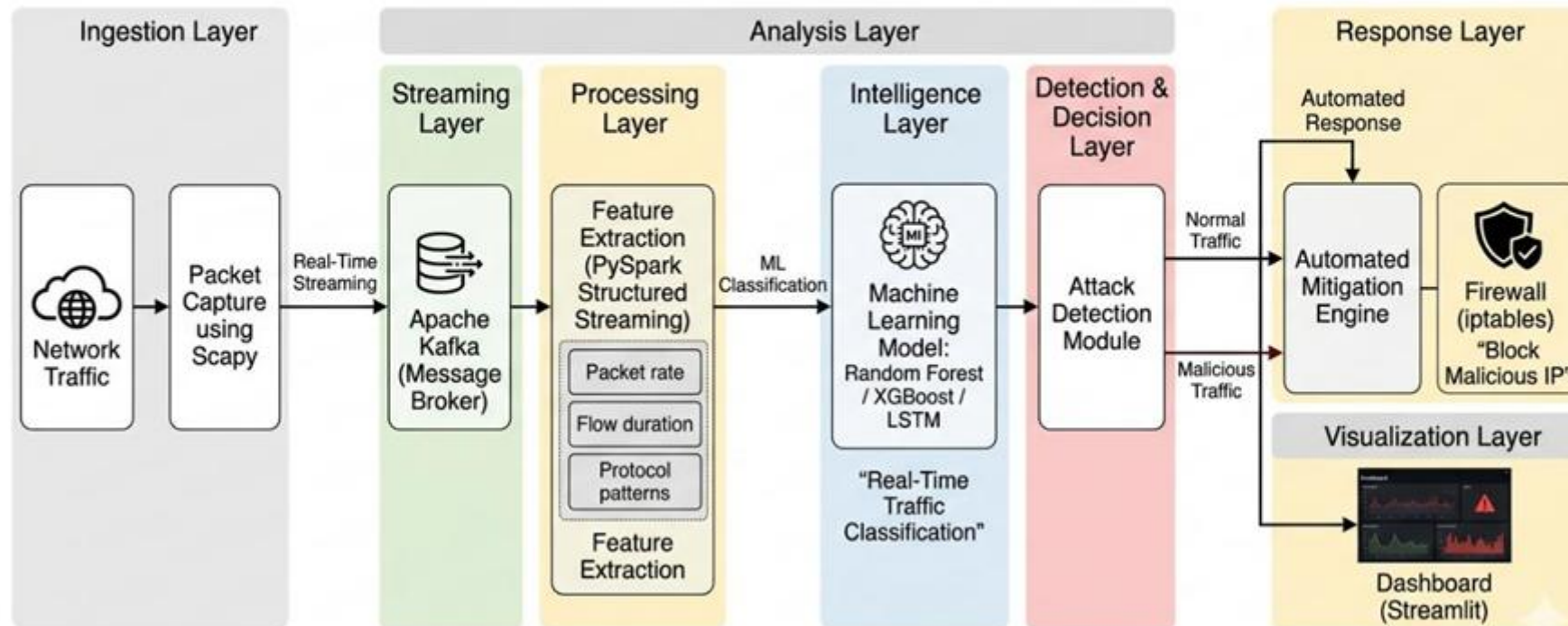
Reaction & Visualization Layer (The Muscle)

- **Stage 5: Autonomous Mitigation:** When a flow is classified as malicious, a Python controller immediately triggers the **"muscle" response**. It injects a "DROP" rule into the **Linux iptables** at the kernel level, silently terminating all further connections from the offending IP address within milliseconds.
- **Stage 6: Visualization & Auditing:** The final stage provides operational visibility through a **Streamlit dashboard**. Security analysts can monitor live throughput, view the geolocation of blocked threats, and access persistent audit logs for forensic analysis and compliance.

Architecture

AutoShield System Architecture

“AutoShield: Real-Time Machine Learning–Driven DDoS Detection and Mitigation System”





Machine Learning Model Performance

- **High Accuracy:** Achieved a peak classification accuracy of **99.85%** on the CICIDS2017 benchmark dataset.
- **Precision Focus:** Maintains a precision score of **0.99** to minimize false positives and prevent blocking legitimate users.
- **Optimized Features:** Used Recursive Feature Elimination (RFE) to prune 80+ features down to the **30 most predictive variables**
- **Ultra-Low Latency:** Delivers near-instant inference with a latency of approximately **1ms per flow**
- **Balanced Training:** Applied **SMOTE** oversampling to correct class imbalances and identify rare attack patterns.
- **Outlier Resistance:** Utilized **RobustScaler** (median/IQR) for normalization to ensure stability during varying traffic conditions.
- **Algorithm Benchmarking:** Outperformed secondary models like **XGBoost** (99.11%) and sequential **LSTM** (~94.6%) in overall precision.



Security Audit & Evaluation

- **Framework Compliance:** System operational protocols are aligned with the **NIST Cybersecurity Framework (CSF)** to ensure robust access control and incident logging.
- **Automated Audit Logging:** Every autonomous mitigation action—including the blocked IP, timestamp, and flow features is recorded in a persistent, patient inspection log.
- **Benchmark Validation:** The system was rigorously evaluated using the industry-standard **CICIDS2017** and **CIC-DDoS2023** datasets to confirm high-fidelity threat detection.
- **Closed-Loop Performance:** Stress testing confirms the architecture satisfies core requirements for autonomous intrusion prevention by achieving a detection-to-block latency of **under 100 milliseconds**



Project Timeline

Phase / Period	Tasks
Phase 1 (Feb)	Requirement Analysis - Defined the project scope for autonomous DDoS mitigation and identified hardware/software needs for network sniffing.
Phase 2 (Feb-Mar)	Environment Setup - Configured Ubuntu 22.04 LTS, Docker containers for Kafka, and initial Python scripts for packet capture using Scapy.
Phase 3 (Mar)	Pipeline Development - Implemented the Kafka producer for real-time traffic ingestion and the PySpark consumer for windowed flow feature extraction.
Phase 4 (Mar)	Model Training & Optimization - Developed Random Forest and XGBoost models using the CICIDS2017 dataset; applied RFE for feature reduction.
Phase 5 (Mar-Apr)	Integration & Mitigation - Connected the ML classifier to the reaction layer to execute automated iptables "DROP" rules at the kernel level.
Phase 6 (Apr)	Visualization & Dashboard - Developed a real-time Streamlit dashboard featuring live throughput analytics, geolocation, and audit logging.
Phase 7 (Apr)	Testing & Deployment - Performed stress testing using attack simulation tools to validate the sub-100ms detection-to-block latency.



Conclusion & Future Scope

- **Autonomous Defense Success:** AutoShield effectively bridges the gap between threat detection and active network defense by achieving an automated detection-to-block latency of under 100 milliseconds.
- **High-Performance Intelligence:** The system demonstrates precision security through 99.85% classification accuracy, utilizing optimized Random Forest models and real-time Spark streaming.
- **Encrypted Traffic Research:** Future scope includes extending detection capabilities to HTTPS-based attacks by analyzing TLS flow statistics without requiring packet decryption.
- **Federated Intelligence:** Plans are in place to implement federated learning across multiple nodes to enable collective threat intelligence while maintaining data privacy.
- **IoT & Edge Adaptation:** Future development aims to adapt the framework for 5G slices and IoT edge nodes using specific datasets like CIC-DDoS2023 for specialized training.



Work Completed

- **Real-Time Packet Ingestion:** Developed a sensory layer using **Scapy** to capture live network traffic in promiscuous mode and stream it into **Apache Kafka**.
- **High-Throughput Streaming Pipeline:** Configured a **Kafka-based buffering system** that acts as a shock absorber to prevent data loss during massive volumetric DDoS spikes.
- **Automated Feature Extraction:** Implemented **PySpark Structured Streaming** to aggregate raw packets into 5-tuple flows and calculate over 20 behavioral metrics, such as protocol entropy and flow duration.
- **Machine Learning Intelligence:** Trained and optimized a **Random Forest model** on the CICIDS2017 dataset, achieving a classification accuracy of **99.85%**.
- **Feature Optimization (RFE):** Utilized **Recursive Feature Elimination** to reduce the feature set from 80+ variables to the 30 most predictive variables to minimize inference latency.
- **Autonomous Mitigation Engine:** Built a Python-driven controller that automatically injects **iptables** rules at the Linux kernel level to block malicious IPs within milliseconds.
- **SOC Monitoring Dashboard:** Developed a real-time **Streamlit dashboard** that provides live throughput analytics, geolocation of blocked threats, and persistent audit logging.
- **System Evaluation & Testing:** Successfully validated the system's performance, achieving a total detection-to-block latency of **under 100 milliseconds**



Thank You